

IV Thrombolysis Before Thrombectomy in Carotid Artery Dissection–Related Large-Vessel Occlusion

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Abstract

Background and Objectives

The benefit of IV thrombolysis (IVT) before thrombectomy in patients with anterior-circulation large-vessel occlusion (LVO) due to carotid artery dissection (CAD) remains uncertain. We aimed to evaluate the safety and efficacy of IVT in this specific population in clinical practice.

Methods

This study included patients from the Carotid Dissection Thrombectomy vs Medical Treatment collaboration and the Endovascular Treatment in Ischemic Stroke registry with anterior-circulation LVO due to CAD who were treated with thrombectomy, with or without prior IVT. The coprimary outcomes were favorable functional outcome at 3 months (modified Rankin Scale 0–2) and successful reperfusion (thrombolysis in cerebral infarction score $\geq 2b$). The safety end point was symptomatic intracranial hemorrhage (sICH). Inverse probability of treatment weighting was applied to adjust for baseline imbalances, and weighted Poisson regression models were used to estimate adjusted risk ratios (aRRs).

Results

A total of 1,091 patients (mean age 53.0 years, 74.0% men) were included, of whom 571 patients received IVT prior thrombectomy and 520 underwent thrombectomy only. IVT before thrombectomy was associated with a higher likelihood of favorable functional outcome (aRR 1.16, 95% CI 1.03–1.30, $p = 0.013$) and successful reperfusion (aRR 1.34, 95% CI 1.18–1.53, $p < 0.001$). No association was observed with sICH (aRR 0.69, 95% CI 0.41–1.16, $p = 0.161$). Subgroup analyses showed that the functional benefit of IVT was pronounced in patients with higher admission NIH Stroke Scale scores (p interaction = 0.039).

Discussion

IVT prior thrombectomy was associated with improved functional outcome and higher reperfusion success in patients with anterior-circulation LVO due to CAD, without an increase in sICH. The association with functional outcome was largely driven by patients presenting with higher admission stroke severity.

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Class of Evidence

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Supplementary Material

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Glossary

AIS = acute ischemic stroke; **aRR** = adjusted risk ratio; **CAD** = carotid artery dissection; **CONCORDIA** = Carotid Dissection Thrombectomy vs Medical Treatment; **ETIS** = Endovascular Treatment in Ischemic Stroke; **iCAO** = internal carotid artery occlusion; **IPTW** = inverse probability of treatment weighting; **IVT** = IV thrombolysis; **LVO** = large-vessel occlusion; **mRS** = modified Rankin Scale; **NIHSS** = NIH Stroke Scale; **sICH** = symptomatic intracranial hemorrhage; **SMD** = standardized mean difference; **TICI** = thrombolysis in cerebral infarction score.

Classification of Evidence

This study provides Class III evidence that IVT before thrombectomy was associated with improved functional outcome and higher reperfusion success in patients with anterior-circulation LVO due to CAD, without an increase in sICH.

Introduction

Carotid artery dissection (CAD) is a rare, but relevant cause of acute ischemic stroke (AIS), particularly in younger adults.^{1,2} While thrombectomy is an established treatment for large-vessel occlusion (LVO) regardless of the underlying cause, the strongest treatment effect in patients with CAD was seen in patients with higher NIH Stroke Scale (NIHSS) scores and tandem occlusions.³⁻⁵ A dissection may complicate thrombectomy by obscuring the true lumen and thereby hindering access to the intracranial vasculature, while associated thrombus formation may promote distal embolization.^{4,6-8} IV thrombolysis (IVT) is frequently administered as part of standard AIS treatment, yet its role in patients with LVO due to CAD is not well defined.⁹ Two recent CAD studies suggest that IVT may be associated with improved functional outcomes and higher rates of home discharge, although data on reperfusion outcomes remain limited.¹⁰⁻¹² The objective of this large collaborative study was to examine the safety and efficacy of IVT administered before thrombectomy in patients with anterior circulation LVO caused by CAD. The primary research questions were whether IVT before thrombectomy improves 3-month functional outcome and reperfusion success and whether it is safe in LVO-related CAD.

Methods

Study Population and Data Collection

The study included patients presenting between 2015 and 2025 with AIS due to an anterior circulation LVO (including tandem, carotid-T, M1, M2 occlusions, and internal carotid artery occlusion [iCAO]) secondary to CAD. All patients were ≥ 18 years of age and were treated with thrombectomy, with or without prior IVT. Data were obtained from the Carotid Dissection Thrombectomy vs Medical Treatment (CONCORDIA) Collaboration and the Endovascular Treatment in Ischemic Stroke (ETIS) registry (ClinicalTrials.gov: NCT03776877). CONCORDIA is an international, multicenter observational study pooling data from 3 nationwide stroke registries: the Austrian Stroke Unit Registry, the German Stroke Registry-Endovascular Treatment, and the Swiss Stroke

Registry.⁵ ETIS is an ongoing prospective, multicenter observational registry conducted in 34 comprehensive stroke centers in France.¹³ Across all registries, data collection, clinical assessments, and documentation were performed by vascular neurologists at the participating stroke centers at admission, discharge, and at 3-month follow-up using standardized definitions for all variables and outcomes. Patients were excluded if IVT status was unclear or if only diagnostic angiography was performed without thrombectomy due to spontaneous or early recanalization. CAD was diagnosed based on vascular imaging (CT angiography, magnetic resonance angiography, or digital subtraction angiography) according to standard clinical practice at the participating centers. Patients presenting in the extended time window, or wake-up strokes receiving IVT based on advanced imaging, were also eligible for inclusion in the study.

Study Outcomes

The coprimary end points were favorable functional outcome at 3 months, defined as a modified Rankin Scale (mRS) score of 0–2, and successful reperfusion after thrombectomy, defined as thrombolysis in cerebral infarction score (TICI) $\geq 2b$. Three-month mRS was assessed either during an in-person visit or by structured telephone interview. The safety end point was symptomatic intracranial hemorrhage (sICH), classified as any intracranial bleeding associated with a neurologic deterioration of at least 4 NIHSS score points occurring within 36 hours after thrombectomy.¹⁴ Follow-up neuroimaging was routinely performed using CT or MRI 22–36 hours after thrombectomy or earlier in case of neurologic deterioration.¹⁵

Statistics

Baseline characteristics were summarized as mean (SD) with range or median with interquartile range for continuous variables and as counts with percentages for categorical variables. Differences between the IVT + thrombectomy and thrombectomy only groups were assessed using the χ^2 test for categorical variables and the Wilcoxon rank-sum test for continuous variables. To assess baseline comparability, standardized mean differences (SMDs) were calculated for all covariates included in the propensity score model.

SMD <0.10 was considered to reflect good balance between groups and <0.20 to indicate slight imbalance. Inverse probability of treatment weighting (IPTW) was applied to account for baseline differences between the 2 groups. Propensity scores were estimated using logistic regression including the following covariates: age, sex, stroke severity (measured by admission NIHSS score), prestroke mRS, arterial hypertension, diabetes mellitus, atrial fibrillation, localization (tandem, carotid-T, M1, M2, iCAO), and onset-to-door time (defined as the time from symptom onset to arrival at the thrombectomy-capable center). Tandem occlusions were defined as combined extracranial high-grade internal carotid artery stenosis (according to North American Symptomatic Carotid Endarterectomy Trial) or occlusion and intracranial LVO, and iCAO as isolated internal carotid artery occlusion, including intracranial internal CADs, but without involvement of an LVO of the circle of Willis.¹⁶ To ensure adequate adjustment in the propensity score model, carotid-T, M1, and M2 occlusions accompanied by extracranial high-grade internal carotid artery stenosis or occlusion were grouped as tandem occlusions for IPTW analyses. Afterward, occlusion sites were coded hierarchically (tandem > carotid-T > M1 > M2 occlusion > iCAO) to ensure mutual exclusivity. Descriptive analyses distinguished between intracranial occlusion site and extracranial carotid involvement. Stabilized IPTW weights were calculated, and extreme weights were trimmed at the 1st and 99th percentiles. For each outcome, crude risk ratios were obtained using unweighted Poisson regression with robust variance estimators. IPTW-adjusted risk ratios (aRRs) and 95% CIs were estimated using weighted Poisson regression models. Sensitivity analyses included doubly robust weighted and covariate-adjusted models with cluster-robust standard errors, alternative propensity score specifications using onset-to-groin time instead of onset-to-door time, models accounting for calendar year of stroke treatment, and an ordinal shift analysis of the full 3-month mRS distribution. Subgroup-specific treatment effects were assessed within each subgroup using IPTW-weighted Poisson regression. Effect modification was evaluated by including interaction terms between IVT and subgroup variables in the weighted models. For NIHSS and onset-to-door time, interactions were examined as continuous variables. Model-based predicted probabilities and absolute risk differences were estimated across admission NIHSS scores. Missing data were handled using a complete-case approach. Differences in denominators reflect outcome specific data availability and the use of complete-case propensity score weighted analyses. Baseline characteristics were compared between patients with available and missing outcome data. Statistical significance was defined as a 2-sided *p*-value <0.05. All statistical analyses were performed using R version 4.4.2 (R Foundation for Statistical Computing, Vienna, Austria).

Standard Protocol Approvals, Registrations, and Patient Consents

The study protocol received approval from the local ethics committee (Sigmund Freud University, No. 1025–2024).

Patient informed consent was not required due to the study's retrospective and anonymized design. This study is reported in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology guidelines.¹⁷

Data Availability

Data supporting the results of this study can be obtained from the corresponding author upon reasonable request and following approval by the national academic review boards.

Results

Study Population

Between 2015 and 2025, CONCORDIA included 517 patients and ETIS 574 patients with anterior circulation LVO caused by CAD, resulting in a total of 1,091 patients (mean age 53.0 [SD 11.3], 74.0% male). Of these, 571 received IVT prior thrombectomy and 520 were treated with thrombectomy only. Three-month mRS scores were available for 495 (86.69%) patients in the IVT + thrombectomy group and for 455 (87.50%) patients in the thrombectomy only group. TICI scores were recorded for 557 (97.55%) patients in the IVT + thrombectomy group and 505 (97.12%) patients in the thrombectomy only group. Baseline characteristics of patients with missing vs available outcomes are presented in eTables 1 and 2.

Baseline Characteristics

Both groups included a higher proportion of male patients, with 436 (76.4%) in the IVT + thrombectomy group and 371 (71.3%) in the thrombectomy only group. Significant differences were observed in onset-to-door time, onset-to-groin puncture time, and in the proportion of M2 and tandem occlusions. Table 1 summarizes the baseline characteristics.

Unweighted and weighted SMDs for baseline covariates are presented in Table 2.

Efficacy and Safety End Points

A favorable functional outcome was observed in 324 (65.45%) patients in the IVT + thrombectomy group and was significantly (*p* = 0.004) more frequent than in the thrombectomy only group with 255 (56.04%) patients (Figure 1). After adjustment, IVT prior thrombectomy remained associated with a higher likelihood of achieving a favorable functional outcome (aRR 1.16, 95% CI 1.03–1.30, *p* = 0.013, Table 3). Ordinal shift analyses of the full mRS distribution showed consistent findings and are listed in eTable 3. A flow diagram for the main analysis is presented in eFigure 1.

A successful reperfusion was achieved significantly (*p* < 0.001) more often in the IVT + thrombectomy group, with 393 (70.56%) patients, compared with the thrombectomy only group with 282 (55.84%) patients. IVT prior thrombectomy was significantly (*p* < 0.001) associated with a higher likelihood of successful reperfusion (aRR 1.34, 95% CI 1.18–1.53).

Table 1 Baseline Characteristics Categorized by IVT + Thrombectomy vs Thrombectomy Only

	IVT + thrombectomy (n = 571)	Thrombectomy only (n = 520)	p Value
Age, y, mean (SD)	53.1 (10.4)	52.9 (12.2)	0.807
Sex, male, n (%)	436 (76.4)	371 (71.3)	0.070
Admission NIHSS, median (Q1–Q3)	15 (10–18)	14 (8–19)	0.322
Prestroke mRS 0–1, n (%)	547 (97.0)	494 (96.3)	0.646
Hypertension, n (%)	176 (31.1)	183 (35.5)	0.144
Diabetes mellitus, n (%)	18 (3.2)	30 (5.8)	0.051
Atrial fibrillation, n (%)	14 (2.6)	20 (3.9)	0.290
Occlusion sites, n (%)			
M1	191 (33.5)	173 (33.3)	0.950
M2	75 (13.1)	93 (17.9)	0.030
Carotid-T	93 (16.3)	97 (18.7)	0.303
Tandem	234 (41.0)	173 (33.3)	0.009
iCAO	57 (10.0)	66 (12.7)	0.158
Onset-to-door time, min, median (Q1–Q3)	85 (60–144)	120 (70–252)	<0.001
Emergent stenting, n (%)	161 (28.2)	167 (32.1)	0.179
Duration of thrombectomy ^a , min, median (Q1–Q3)	53 (33–83)	55 (34–96)	0.088
Onset-to-groin puncture time, min, median (Q1–Q3)	220 (162–300)	272 (180–450)	<0.001

Abbreviations: iCAO = internal carotid artery occlusion—isolated internal carotid artery occlusion without intracranial large vessel occlusion involving the circle of Willis; IVT = IV thrombolysis; mRS = modified Rankin Scale; NIHSS = NIH Stroke Scale.

^a Time from arterial puncture to the final angiographic assessment.

Regarding the safety end point, sICH occurred in 46 (8.06%) patients in the IVT plus thrombectomy group and in 43 (8.27%) patients in the thrombectomy only group ($p = 0.986$). IVT prior thrombectomy was not associated with sICH (aRR 0.69, 95% CI 0.41–1.16, $p = 0.161$). Effect estimates from sensitivity analyses were consistent with the primary IPTW analyses and are listed in eTable 4.

Subgroup Analyses According to the Stroke Severity, Onset-to-Door Time, Occlusion Sites, and Emergent Stenting

When modeled as a continuous variable, admission NIHSS significantly modified the association between IVT before thrombectomy and favorable functional outcome (p interaction = 0.039). The relative association of IVT with favorable outcome increased with higher NIHSS (aRR per NIHSS score 1.02, 95% CI 1.00–1.04). Model-based predicted probabilities and absolute risk differences indicated that the absolute association of IVT increased with higher admission NIHSS (Figure 2).

In contrast to stroke severity, no significant effect modification of the association between IVT before thrombectomy and favorable functional outcome was observed by onset-to-door time (p interaction = 0.210) or by occlusion site (p

interaction = 0.679) (Figure 3). IVT prior thrombectomy was not associated with favorable functional outcome in patients with tandem (aRR 1.17, 95% CI 0.95–1.43), carotid-T (aRR 1.19, 95% CI 0.88–1.60), M1 (aRR 1.24, 95% CI 0.98–1.55), M2 occlusions (aRR 1.19, 95% CI 0.89–1.59), or iCAO (aRR 1.09, 95% CI 0.81–1.48). Similarly, there was no effect modification by emergent carotid artery stenting (p for interaction = 0.770). The estimated treatment effects were similar in patients undergoing emergent stenting (aRR 1.13, 95% CI 0.95–1.36) and in those without emergent stenting (aRR 1.17, 95% CI 1.01–1.36).

In analyses accounting for admission NIHSS and onset-to-door time as continuous variables, no significant effect modification of the association between IVT prior thrombectomy and successful reperfusion was observed by admission NIHSS (p interaction = 0.193) or onset-to-door time (p interaction = 0.069).

In occlusion-site subgroup analyses, the association between IVT prior thrombectomy and successful reperfusion differed significantly by occlusion site (p interaction = 0.028) (Figure 4). IVT prior thrombectomy was associated with successful reperfusion in tandem (aRR 1.18, 95% CI 1.01–1.38, $p = 0.037$) and M1 occlusions (aRR 2.35, 95% CI

Table 2 SMDs Before and After IPTW

	Unweighted SMD	Weighted SMD ^a
Age	0.045	0.001
Sex	0.125	0.009
Prestroke mRS	0.123	0.015
Admission NIHSS	0.165	0.011
Hypertension	0.055	0.005
Diabetes mellitus	0.140	0.006
Atrial fibrillation	0.045	0.011
Occlusion sites		
M1	0.030	0.004
M2	0.062	0.004
Carotid-T	0.017	0.002
Tandem	0.106	0.011
iCAO	0.107	0.011
Onset-to-door time	0.214	0.106

Abbreviations: iCAO = internal carotid artery occlusion—isolated internal carotid artery occlusion without intracranial large vessel occlusion involving the circle of Willis; IPTW = inverse probability of treatment weighting; mRS = modified Rankin Scale; NIHSS = NIH Stroke Scale; SMD = standardized mean difference.

^a Displayed values reflect covariate balance after IPTW, SMD <0.10 indicates good balance, values between 0.10 and 0.20 reflect minor residual imbalance.

1.58–3.49, $p < 0.001$), whereas no significant associations were found in carotid-T (aRR 1.20, 95% CI 0.97–1.50, $p = 0.100$), M2 occlusions (aRR 0.79, 95% CI 0.40–1.54, $p =$

0.486), or iCAO (aRR 1.32, 95% CI 0.88–1.97, $p = 0.174$). Similarly, effect modification by emergent carotid artery stenting was observed (p interaction = 0.020). IVT was associated with successful reperfusion in patients without emergent stenting (aRR 1.49, 95% CI 1.27–1.76, $p < 0.001$), but not in those undergoing emergent stenting (aRR 1.09, 95% CI 0.89–1.34, $p = 0.393$).

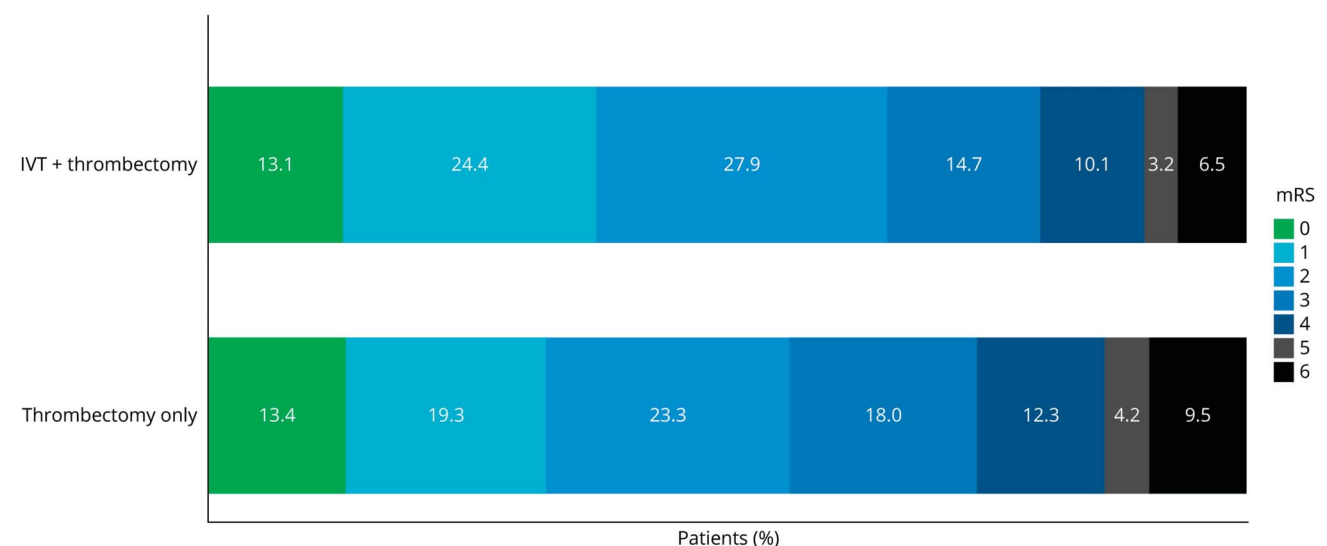
Across occlusion-site and emergent carotid stenting subgroup analyses, IVT before thrombectomy was not associated with sICH. No effect modification of the association between IVT before thrombectomy and sICH was observed by admission NIHSS (p interaction = 0.397) or onset-to-door time (p interaction = 0.174).

Classification of Evidence

This study provides Class III evidence that IVT before thrombectomy was associated with improved functional outcome and higher reperfusion success in patients with anterior-circulation LVO due to CAD, without an increase in sICH.

Discussion

This multicenter study provides 3 central findings: (1) IVT prior thrombectomy was independently associated with 3-month favorable functional outcome and angiographic reperfusion in patients with anterior circulation LVO due to CAD; (2) IVT prior thrombectomy was not associated with an increased risk of sICH, and therefore showed no safety concerns; and (3) the treatment effect on functional outcome was most pronounced in patients presenting with higher stroke severity at admission.

Figure 1 Distribution of 3-Month mRS by Treatment Group

IVT = IV thrombolysis; mRS = modified Rankin Scale.

Table 3 Efficacy and Safety Outcomes by IVT + Thrombectomy vs Thrombectomy Only

Outcome	IVT + thrombectomy, n (%) ^a	Thrombectomy only, n (%) ^a	RR (95% CI)	p Value	aRR (95% CI)	p Value
mRS 0–2	245 (69.4)	192 (61.5)	1.17 (1.05–1.30)	0.003	1.16 (1.03–1.30)	0.013
TICI ≥2b	274 (69.0)	168 (49.1)	1.26 (1.15–1.39)	<0.001	1.34 (1.18–1.53)	<0.001
sICH	26 (6.4)	30 (8.6)	0.97 (0.65–1.45)	0.898	0.69 (0.41–1.16)	0.161

Abbreviations: aRR = adjusted risk ratio; IPTW = inverse probability of treatment weighting; mRS = modified Rankin Scale; RR = risk ratio; sICH = symptomatic intracranial hemorrhage; TICI = thrombolysis in cerebral infarction.
^a Denominators differ across outcomes due to outcome specific data availability and the use of complete case propensity score weighted analyses. The exact numbers of patients included in each analysis are shown in eFigure 1. Effect estimates are presented as unadjusted RR from the original cohort and IPTW-adjusted RR from the weighted population.

The use of IVT in CAD has long been debated. In clinical practice, one of the arguments for IVT is that cervical dissections often generate thrombus material that may embolize distally, and IVT is assumed to help dissolve these clots.^{8,18} However, this assumption has not been backed by strong evidence. The 2021 European Stroke Organization guidelines

Figure 2 Association of IVT Prior Thrombectomy With Favorable Functional Outcome (3-Month mRS 0–2) Across Admission NIHSS Scores

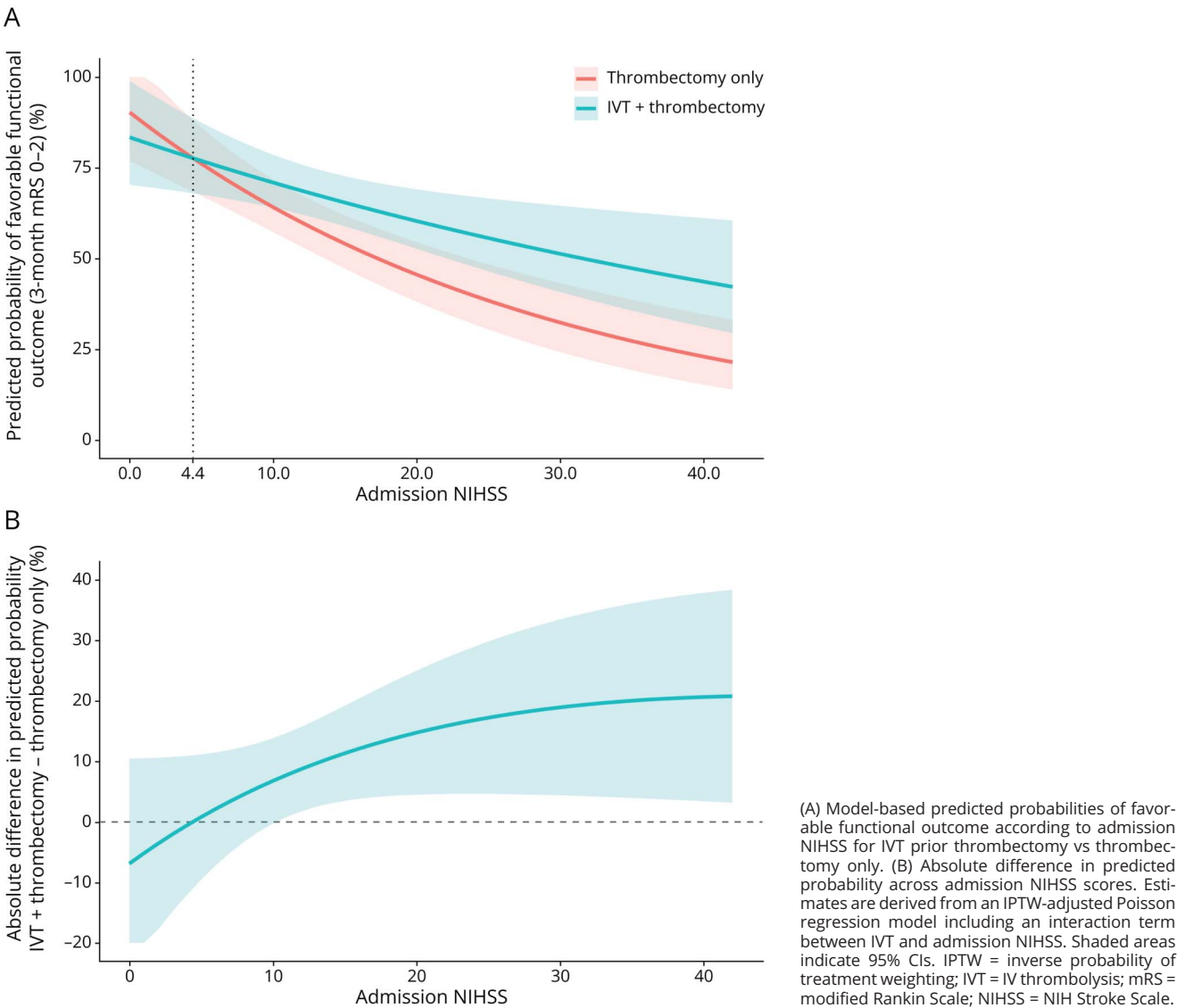
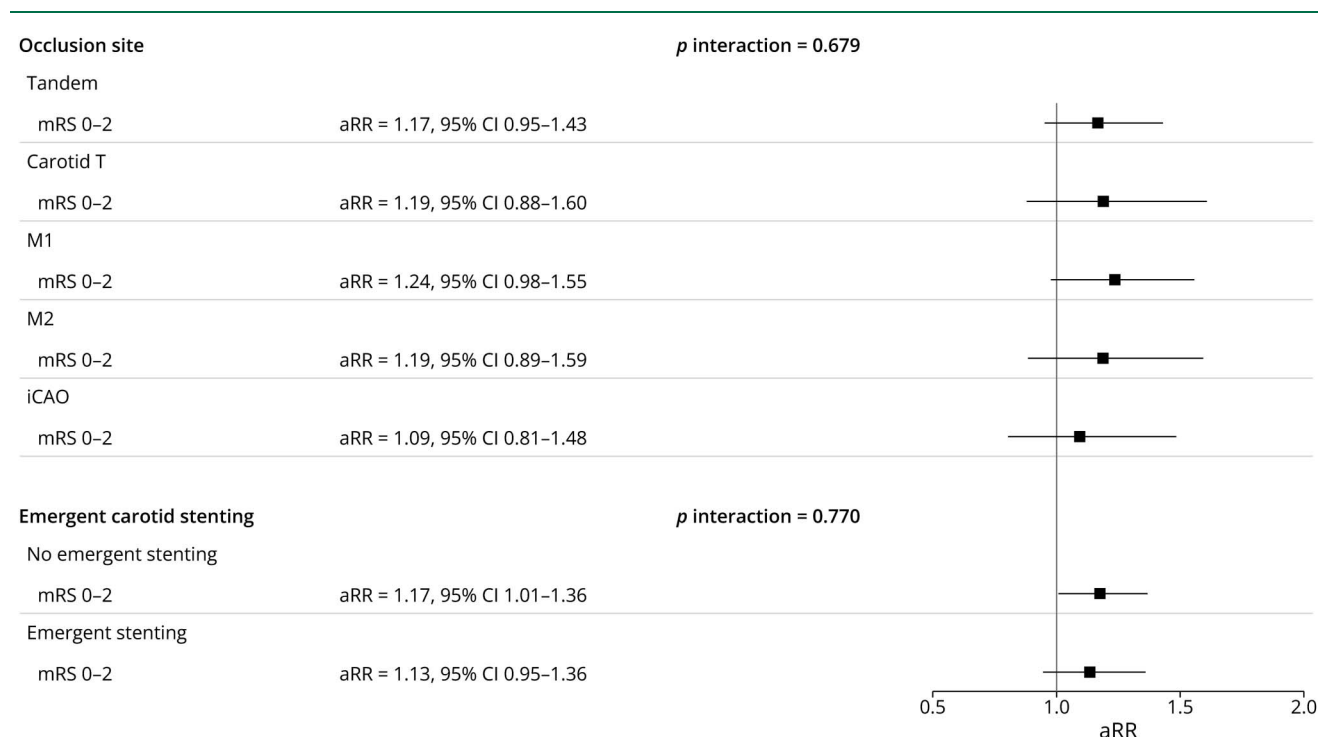


Figure 3 Subgroup Analyses of Favorable Functional Outcome (3-Month mRS 0–2) for IVT + Thrombectomy vs Thrombectomy Only



Effect estimates are presented as IPTW-adjusted RR from the weighted population. aRR = adjusted risk ratio; iCAO = internal carotid artery occlusion—isolated internal carotid artery occlusion without intracranial large vessel occlusion involving the circle of Willis; IPTW = inverse probability of treatment weighting; mRS = modified Rankin Scale; RR = risk ratio.

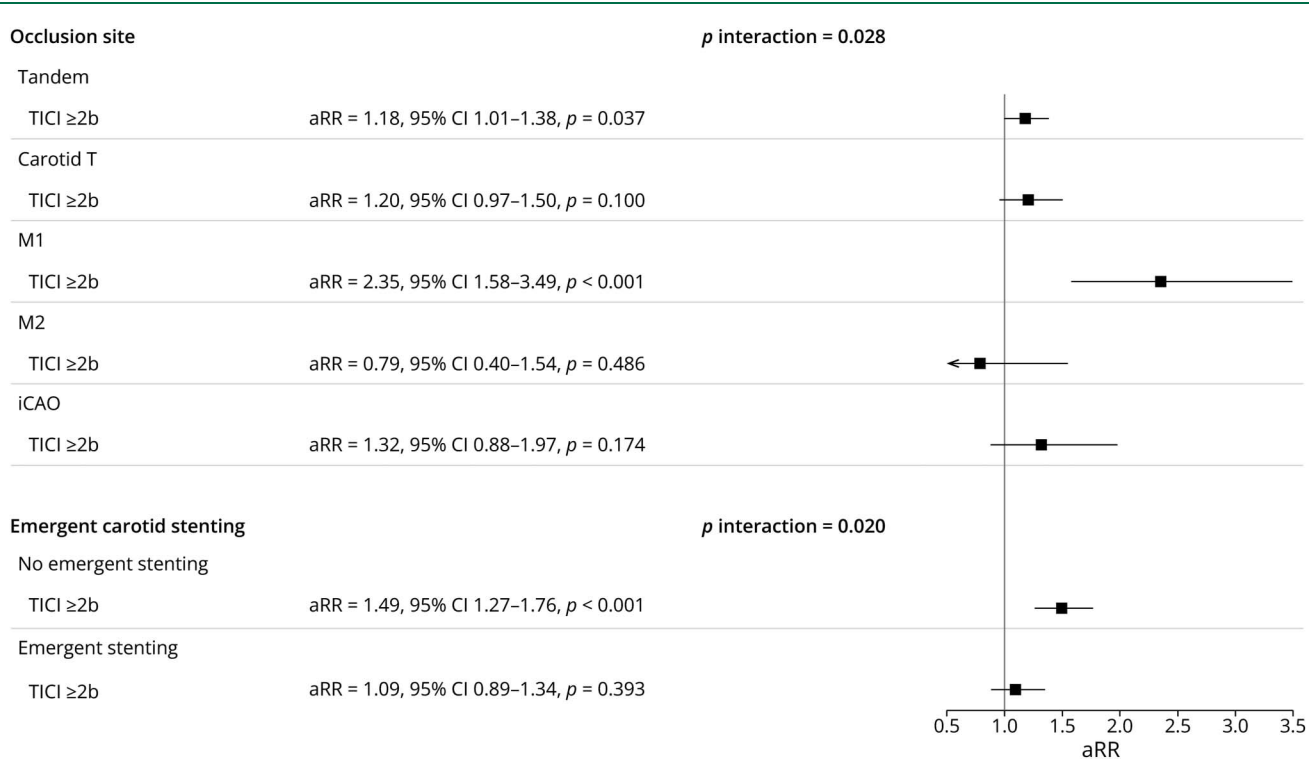
therefore describe IVT for CAD as low-quality evidence, mainly because earlier studies relied on subgroup analyses, which did not show a clear functional benefit.⁹ Further, findings from other stroke etiologies cannot be directly transferred to CAD. Patients with CAD are younger, more often male, have fewer vascular risk factors, and less prestroke disability than the typical AIS population.^{1,19} More recently, with STOP-CAD and a U.S. administrative database analysis, 2 larger studies revisited this question.^{10,11} STOP-CAD included 1,653 patients and reported better functional outcomes in patients with CAD treated with IVT compared with best medical treatment and without an increased risk of sICH. The US study similarly found that IVT was associated with a higher likelihood of discharge to home, particularly in patients with more severe neurologic deficits. Similar findings have been reported for other stroke etiologies and are consistent with our observation in patients with CAD-related LVO that the association between IVT before thrombectomy and favorable functional outcome was more pronounced at higher admission NIHSS scores, whereas no clear association was observed at lower NIHSS scores.^{20–22} Although the use of IVT and the benefit of thrombectomy in patients with low NIHSS scores and LVO remain controversial, our data do not allow firm conclusions in this subgroup.²³ In analyses stratified by admission NIHSS, no association was observed between admission NIHSS and the occurrence of sICH. The overall rate of sICH was approximately 8% in both treatment

groups and comparable to other thrombectomy cohorts.^{24–26} In general, IVT was not associated with sICH, which indicates that IVT prior thrombectomy is generally safe in CAD related LVO.

In our cohort, onset-to-door time did not significantly modify the association of IVT before thrombectomy with either reperfusion success or favorable functional outcome. These findings suggest that the association of IVT with both outcomes was consistent across different time windows. However, owing to the exploratory nature of subgroup analyses, no solid conclusion regarding the time-dependent effects of IVT can be drawn.^{27–30}

When stratified by occlusion site, IVT before thrombectomy was most strongly associated with successful reperfusion in patients with M1 and tandem occlusions, whereas no consistent association with favorable functional outcome was observed. This is in line with previous observational studies, which have suggested a synergistic effect of IVT and thrombectomy, potentially through clot modification and dissolution of distal thrombus fragments.^{31–33} However, randomized controlled trials have not shown a functional benefit of IVT in tandem occlusions; thus, the difference in findings may eventually relate to selection bias and residual confounding inherent to observational analyses.³⁴ In addition, Improving Reperfusion Strategies in Acute Ischemic Stroke meta-

Figure 4 Subgroup Analyses of Successful Reperfusion (TICI ≥2b) for IVT + Thrombectomy vs Thrombectomy Only



Effect estimates are presented as IPTW-adjusted RR from the weighted population. aRR = adjusted risk ratio; iCAO = internal carotid artery occlusion—isolated internal carotid artery occlusion without intracranial large vessel occlusion involving the circle of Willis; IPTW = inverse probability of treatment weighting; mRS = modified Rankin Scale; RR = risk ratio; TICI = thrombolysis in cerebral infarction.

analysis did not differentiate between CAD related and atherosclerotic tandem occlusions and is likely to reflect a predominantly atherosclerotic population. CAD represents a distinct pathophysiology, which may partly explain differences in the observed effect of IVT in our study. Furthermore, patients with CAD tend to be younger and have fewer vascular comorbidities than those with atherosclerotic disease, which may further contribute to a higher likelihood of favorable functional recovery.³⁵

IVT prior thrombectomy was further associated with higher TICI scores only in patients who did not undergo emergent stenting. This finding may be eventually explained by the fact that emergent carotid stenting typically achieves high rates of successful recanalization on its own, which may reduce the potential for IVT to provide additional benefit.²⁶

Several limitations of this study must be acknowledged. First, incomplete information on admission pathways (direct referrals vs secondary transfers) represents a potential source of bias. This limitation is of particular relevance in patients with CAD, in whom treatment decisions regarding IVT and thrombectomy may differ substantially depending on the site of initial presentation. Furthermore, reasons for non-administration of IVT and imaging-based markers such as Alberta Stroke Program Early CT Score or infarct core volume were not consistently available across all registries and

could therefore not be included in the analyses, which may have contributed to confounding. Although onset-to-door time was accounted for in the analyses, residual imbalance suggests that this factor might still represent eventual source of bias. Nonetheless, the consistency of results across several complementary analyses supports the conclusion that the observed associations in our study are unlikely to be explained by differences in presentation times. A further limitation is that CAD diagnoses were based on local vascular imaging assessment without centralized validation. As with all observational studies, bias by indication cannot be ruled out, and residual confounding may persist, although IPTW and robust Poisson regression were applied to reduce baseline differences. Although most covariates achieved good balance after weighting, a few variables showed small residual SMD values, which should be considered when interpreting the results. Incomplete data on covariates and follow-up visits, particularly missing 3-month mRS scores and fewer missing TICI scores, may have introduced some uncertainty into the outcome analyses. Comparisons between patients with available and missing data indicated that missingness was likely not completely at random, which suggest that a complete-case approach may be subject to selection bias. In addition, data on antithrombotic therapy for secondary prevention were not consistently available and may have influenced recurrent events, ICH risk and functional outcomes. Given the limited sample sizes within subgroups, these results should be

interpreted with caution and regarded as hypothesis generating only.

Despite limitations, the study benefits from a large multinational prospective cohort, carefully collected registry data and a setting that closely reflects routine clinical practice.

In patients with anterior circulation LVO due to CAD, IVT before thrombectomy was safe and was associated with improved functional outcome and higher reperfusion success. The association with functional outcome was largely driven by patients presenting with higher stroke severity at admission.

Author Contributions

J.F. Hotz: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. S. Poli: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design. M.P. Giannakakis: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. L. Kaindl: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. D. Miksova: analysis or interpretation of data. J. Fladt: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. J. Kaesmacher: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. Z. Kulcsár: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. G. Saliou: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. S. Richard: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. G. Marnat: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. I. Sibon: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. R. Pop: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. V. Wolff: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. A. Consoli: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. J. Mbroh: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. F. Hennersdorf: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. B. Bender: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. J. Fiehler: drafting/revision of the manuscript for content, including medical writing for content; major role in the

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Appendix Coinvestigators

Coinvestigators are listed at [Neurology.org/N](https://www.neurology.org/N).

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